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CHAPTER 1

THEORETICAL BACKGROUND

1.1 Grammatical gender in Polish

There are multiple options of defining grammatical genders in Polish. The one chosen for this study was proposed by Mańczak (1956), where he differentiates between 5 genders:

- masculine personal
- masculine animate
- masculine inanimate
- feminine
- neuter

Saloni (1976) proposed an even more elaborate system, where he talked about 9 genders (instead of one neuter there were two and additionally there were three genders for plurale tantum nouns). Such a system is, however, not necessary for the current analysis, therefore we will adopt the 5 genders proposed by Mańczak. For ease of reference we will use for them the names that Saloni used, which are: M1 (masculine personal), M2 (masculine animate), M3 (masculine inanimate), F (feminine) and N (neuter).

The difference between the genders can be observed when looking at appropriate forms of adjectives in the accusative case. Adjectives agree their genders with nouns and the accusative case provides the most diversity in terms of gender forms.

The example presented here comes from Mańczak (1956). There will be 5 nouns and one adjective for all of them used as an example: *mężczyzna* (meaning 'man'), *pies* (meaning 'dog'), *stół* (meaning 'table'), *kobieta* (meaning 'woman'), *dziecko* (meaning 'child') and *dobry* (meaning 'good').

dobrego	mężczyznę	dobrych	mężczyzn	masculine personal
	psa	dobre	psy	masculine animate
dobry	stół		stoły	masculine inanimate
dobrą	kobietę		kobiety	feminine
dobrze	dziecko		dzieci	neuter

1.2 Previous work on hybrid agreement

1.3 Polish nouns with hybrid gender

1.3.1 Socially gendered nouns

1.3.2 Profession nouns

1.3.3 Depreciative forms of M1 nouns

1.4 Hypothesis

CHAPTER 2

METHODOLOGY

2.1 Agreement sources

The corpus used for this study was the 300M subcorpus of the National Corpus of Polish (Przepiórkowski, 2012), which was parsed using Trankit (Nguyen, Lai, Veyseh, & Nguyen, 2021). The dependency trees (conforming to the Universal Dependency standard, henceforth UD) obtained this way were searched for words, which could trigger hybrid agreement: profession nouns, socially gendered nouns and depreciative forms of M1 nouns. The subsections below describe how those nouns were chosen.

2.1.1 Profession nouns

Nouns in this category came from two studies about feminitives in Polish. The first was a survey reported by Szpyra-Kozłowska (2019), where participants were asked to create feminitives from 50 masculine nouns, that in modern Polish do not have feminine counterparts at all or those counterparts are rarely used. For each noun participants had the option to give no answer. The second study was an experiment conducted by Szpyra-Kozłowska and Laidler (2022), where they tested whether certain feminitives are less accessible by asking participants to try to pronounce them. They chose 20 feminitives, which are formed from masculine nouns that end in consonant clusters such as /kt/, /tr/ or /rg/. According to Klemensiewicz (1957) adding the -ka suffix, the function of which can be to create the feminitive, makes the word difficult to pronounce. Those two studies give a list of 60 nouns in total, for which it is said to be difficult to form feminine counterparts. It is therefore likely that, if there was a female referent of a noun from this list, the masculine noun would be used, leading to hybrid agreement.

Both of the studies determine in some way which of the tested nouns have less available feminitives than others: in Szpyra-Kozłowska and Laidler (2022) those are the nouns for which there are the highest error rates in pronunciation, whereas in Szpyra-Kozłowska (2019) – the nouns for which the most people refused to propose a feminitive. Taking this approach has, however, two downsides: first, it requires an arbitrary threshold that

separates the nouns that have an easily formed feminine from the nouns for which creating the feminine counterpart is more difficult. Second, this approach significantly limits the number of nouns for analysis. Considering this, the current study used all of the nouns from the two studies on Polish femininitives.

For ease of reference, all of those nouns are called *profession* nouns, despite not all of them actually referring to professions. Meanings of all nouns are presented in Table 2.1 in the Appendix.

2.1.2 Socially gendered nouns

The nouns for this category were found using the Polish Academy of Sciences Great Dictionary of Polish available online.¹ The advanced search option allowed to search for nouns in the thematic category HUMAN AS A PHYSICAL BEING: HUMAN PHYSICALITY: SEX. This search yielded 586 results, which then had to be manually filtered to find the nouns that only refer to humans (and not, for example, abstract nouns or nouns referring to body parts) and that had incongruent grammatical and semantic gender. The 19 nouns selected this way can be found in Table 2.1 in the Appendix.

2.1.3 Depreciative forms

For this category no list of nouns was created, since the corpus was annotated and it was possible to look up the depreciative forms by checking the words' morphological description. It is possible to create depreciative forms of surnames, however these were excluded from the analysis, because there was no way of determining the referents and therefore no way of ensuring that there is hybrid agreement. Surname *Machała* found in the corpus can serve as an example: let us say that the *Machała* family bought a new car. Assuming the family consists of at least one man, the speaker can use both (1a) and (1b) to convey that information, depending on their attitude they have towards the situation: (1a) if the speaker is neutral or positive and (1b) if they are negative. However, if there are only women in the family, the speaker can only use (1b). Without the context it is unclear whether *Machały* in (1b) is used as a depreciative or simply to refer to women and therefore we cannot decide if this is an example of hybrid agreement.

- (1) a. Machał-owie kupi-li nowe auto.
Machała-pl.nom.m1.ndepr buy-3pl.m1.past new car
- b. Machał-y kupi-ły nowe auto.
Machała-pl.nom.m1.depr buy-3pl.f.past new car
“Machałas bought a new car.”

Morphological analyzer Morfeusz (Kieraś & Woliński, 2017) can mark forms as surnames which made it easy to exclude them from the analysis.

¹<https://wsjp.pl/>

2.1.4 Finding agreement sources in the corpus

The code used for extracting corpus data in this study can be found in a GitHub repository.² The general idea of how the code works is presented in this chapter.

After choosing the nouns to be searched in the corpus, it was necessary to find all of their forms and the forms' morphological descriptions, which was done using the morphological analyzer Morfeusz. First the corpus was searched for occurrences of nouns, which could be sources of hybrid agreement. Those nouns were saved if they were in the list of gendered or profession noun forms or if they were marked as depreciative in their morphological description. To ensure that no homonyms were accidentally saved for later analysis, the morphological description of a found word was compared with the one given by Morfeusz. This way, for example, the word *premiera* was excluded if it was an instance of the lexeme *premiera*, meaning *premiere* but not if it was an instance of the lexeme *premier*, meaning *prime minister*, in the singular accusative form.

Once it was sure that the right noun was found, the dependency tree was searched for words that can agree their gender with that of the noun. This was the head of the noun (which could be an example of predicative agreement), adjectives attached to the noun (which could be examples of attributive agreement), and relative pronouns referring to the noun. Based on the UD annotation guidelines, all dependents of the word were searched for a dependency label with a subtype :RELCL, which in the UD annotation scheme mark the head of the relative clause. The relative pronoun was attached to the head of the relative clause and had a UD feature PRONTYPE=REL.³ Once the possible agreement targets were found, the data could be used to look for occurrences of gender agreement.

2.2 Listing agreement occurrences

To analyse hybrid gender agreement in the found examples, it is necessary to mark the alternative to the grammatical gender of each agreement source. For all profession nouns that alternative gender was feminine, because the only chance of hybrid agreement for those noun is when the gender of the referent is feminine. For socially gendered nouns the alternative gender was assigned depending on the lexeme.

Once the alternative gender was determined, it was necessary to exclude plural nouns for which the gender discrepancy was between feminine and neuter. This is because for all types of agreement tested here, the plural feminine and plural neuter forms of targets of agreement are syncretic, therefore there is no hybrid agreement in those cases. This happens, for instance, in (2), where in singular ((2a) and (2b)) there is a difference between the feminine and neuter forms of the word *młoda*. In plural (2c), however, the form is the same for both neuter and feminine, so hybrid agreement cannot be observed. Examples like this were therefore excluded.

²<https://github.com/bmagdab/hybrid-agreement>

³https://universaldependencies.org/workgroups/newdoc/relative_clauses.html

- (2) a. młod-e dziewczę
 young-sg.n.nom girl.sg.n.nom
- b. młod-a dziewczę
 young-sg.f.nom girl.sg.n.nom
- c. młod-e dziewczę-ta
 young-pl.nom girl-pl.n.nom
 “young girl”

The next step was to find occurrences of different types of agreement. The predicative agreement of a given noun was found in its head on three conditions: first, dependency label between the noun and the head had to be *NSUBJ*, which is the UD label for nominal subjects. Second, the head had to be a verb that inflects for gender, which is only true for verbs annotated as past participles,⁴ active and passive participles. Third, the noun has to be in the nominative case. This condition helps filter out situations where a numeral phrase is in subject position, because then the verb does not agree with the subject – it is in 3rd person, singular and neuter form, regardless of the noun. Therefore, predicative agreement cannot be observed in those situations at all and they should be excluded.

The search was simpler with attributive and relative pronoun agreement. The only requirement for attributive agreement was that the word is an adjective, which was verified by looking for the tag *ADJ* in the morphological description of the dependent. As for relative pronouns, all of the ones found earlier were included in the analysis, unless they did not inflect for gender (which was the case, for example, for the word *gdzie*, meaning *where*).

The sentence in (3) serves as an example of the process of extracting agreement described in this chapter. The word, which could trigger hybrid agreement (but here does not, since the referent is male) is *premierem*. There are two words in the sentence that agree with this word in terms of gender: *jedynym* and *który*. All of the information saved for each instance of agreement is shown below.

- (3) Nie jest prawd-ą, że premier Jerzy Buzek
 not be.3sg.pres truth.f.sg-inst that prime.minister.sg.m1.nom Jerzy Buzek
 jest jedyn-y-m- premier-em, któr-y
 be.3sg.pres only-m1-inst-sg prime.minister.m1-sg.inst which-sg.nom.m1
 wystąpi-ł w Senac-ie.
 appear-3sg.m1.past in senate.m2-sg.loc
 “It’s not true, that prime minister Jerzy Buzek is the only one that has appeared in the Senate.”

⁴The annotation scheme used for NKJP uses the past participle label not only for past participles, but also for parts of past tense and conditional mood forms (Przepiórkowski, 2012).

		Occurrence 1	Occurrence 2
source of agreement	form	premierem	
	lexeme	premier	
	morphological description	subst:sg:inst:m1	
	grammatical gender	m1	
	other gender	f	
agreement target	form	jedynym	który
	morphological description	adj:sg:inst:m1:pos	adj:sg:nom:m1:pos
	gender	m1	m1
type of agreement		attributive	relative pronoun

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APPENDICES

Table 2.1: Socially gendered and profession nouns in Polish and their translations or definitions in English

Noun in Polish	Translation or definition in English	Category
adiunkt	someone who holds a PhD title and is employed at a scientific institution	profession
architekt	architect	profession
babochłop	a woman that resembles a man with her physicality, looks and behavior	gendered
babol	a woman that looks or behaves in a way that causes aversion in the speaker	gendered
babon	a woman, about whom the speaker feels negatively	gendered
babsko	a woman, about whom the speaker feels negatively	gendered
babsztyl	a woman, about whom the speaker feels negatively	gendered
babus	a woman, about whom the speaker feels negatively	gendered
burmistrz	mayor	profession
chirurg	surgeon	profession
chłopaczysko	a tall or big young man	gendered
chłopobaba	a man that resembles a woman with his physicality, looks and behavior (can also mean a woman that resembles a man)	gendered
ciota	a homosexual man (can also refer to a woman about whom the speaker feels negatively)	gendered
demiurg	demiurge	profession
dramaturg	playwright	profession
drań	scoundrel	profession
dupek	asshole	profession
dureń	fool	profession
dziekan	dean	profession

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Table 2.1: Socially gendered and profession nouns in Polish and their translations or definitions in English (Continued)

Noun in Polish	Translation or definition in English	Category
dziewczątko	a girl, whom the speaker likes or to whom they feel compassion	gendered
dziewczę	a young girl	gendered
dziewuszyisko	a girl who causes aversion or contempt in the speaker	gendered
ekspert	expert	profession
elekt	person elected in a specified post, but has not yet entered office	profession
foniatra	phoniatician	profession
ftyzjatra	pulmonologist	profession
geometra	geometer	profession
geriatra	geriatrician	profession
ginekolog	gynecologist	profession
głupek	fool	profession
herszt	ringleader	profession
inżynier	engineer	profession
jaskiniowiec	caveman	profession
jeniec	captive	profession
jeździec	rider	profession
kierowca	driver	profession
kobiecisko	expressively about a woman, augmentative, or to refer to a big woman	gendered
kobieciątko	expressively about a woman, diminutive	gendered
kobieton	a woman that resembles a man with her physicality, looks and behavior	gendered
ksiądz	priest	profession
lump	tramp	profession
magister	master (holder of a university degree)	profession
majster	foreman	profession
metalurg	metallurgist	profession
minister	minister	profession
murarz	bricklayer	profession

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Table 2.1: Socially gendered and profession nouns in Polish and their translations or definitions in English (Continued)

Noun in Polish	Translation or definition in English	Category
muzykolog	musicologist	profession
mędrzec	wise man	profession
naukowiec	scientist	profession
nieuk	dunce	profession
nurek	diver	profession
pacholę	a boy	gendered
pajac	clown	profession
pannisko	expressively about a young woman, augmentative	gendered
pastor	pastor	profession
pasztet	an ugly and unattractive woman	gendered
pediatra	pediatrician	profession
petent	applicant	profession
polityk	politician	profession
porucznik	lieutenant	profession
powstaniec	insurgent	profession
prefekt	prefect	profession
premier	prime minister	profession
prezydent	president	profession
proboszcz	parish priest	profession
psychiatra	psychiatrist	profession
półkownik	colonel	profession
rektor	person in charge of a university	profession
sierżant	sargeant	profession
starosta	highest administrative officer of a second-level unit of local government	profession
subiekt	salesman	profession
szklarz	glazier	profession
szpieg	spy	profession
szubrawiec	rascal	profession
upiór	phantom	profession

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Table 2.1: Socially gendered and profession nouns in Polish and their translations or definitions in English (Continued)

Noun in Polish	Translation or definition in English	Category
wamp	a confident, misterious, provocative and attractive woman who seduces men	gendered
widz	audience member	profession
wójt	highest administrative officer of a basic-level unit of local government	profession
zbieg	fugitive	profession